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Tsl7tk:
Smart Automotive Services Platform

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Introduction

1. Topic Overview & Importance

TSL7TK is a dedicated mobile utility platform designed to bridge the operational gaps within the Egyptian automotive service and spare-parts ecosystem. In urban centers like Cairo, vehicle owners frequently face significant friction when handling unexpected breakdowns, locating reliable mechanics, or sourcing genuine spare parts. Drivers often encounter fragmented marketplace data, opaque pricing structures, unverified service providers, and uncertain parts availability, leading to extended vehicle downtime and increased repair costs.

The core importance of **TSL7TK** lies in deploying a localized, location-based mobile solution using **Flutter**, **Dart**, and **PostgresSQL**, coupled with geolocation services and a Diagnostic Decision Support System (DSS) integrated into an AI-powered chatbot. By enabling a unified journey from initial breakdown diagnosis and mapping nearby providers to spare-parts availability checks, roadside SOS requests, and parts delivery, the platform enhances service transparency. Furthermore, it empowers drivers with automated maintenance tracking and fuel logs, while providing service centers and vendors with digital tools to update real-time stock, pricing, and availability.

2.Egypt Vision 2030 Contribution & Sustainable Development Goals

Egypt Vision 2030 is built on three dimensions: economic, social, and environmental. Tsl7tk contributes to each.

2.1 Economic dimension: digital transformation, innovation, and small-business growth

Tsl7tk brings independent mechanics, workshops, and parts sellers into a digital marketplace. This gives them visibility, a wider customer base, and tools to manage their own profiles, listings, prices, and stock. It also supports Egypt's move toward a digital economy and a technology-driven startup ecosystem. The relevant features are provider profiles and listings, request management, parts publishing, and provider verification.

2.2 Social dimension: social justice, quality of life, and transparency

The platform makes service and parts pricing, availability, and quality more transparent for drivers. Moderated reviews linked to completed requests build trust and reduce reliance on word of mouth. Roadside assistance and the diagnostic chatbot give drivers faster and more reliable support when problems arise. The relevant features are ratings and moderated reviews, update timestamps, roadside services, and the chatbot with its diagnostic DSS.

2.3 Environmental dimension: sustainable resource use and environmental protection

Maintenance reminders and fuel-consumption tracking encourage drivers to keep vehicles well maintained and efficient. Access to both new and used parts supports reuse and extends the life of vehicles and components. The relevant features are maintenance reminders, the fuel log, and used-parts listings.

The project also supports the government's wider digital transformation agenda by moving a traditionally offline, trust-based market onto a structured digital platform.

Alignment with the Sustainable Development Goals (SDGs)

SDG 3: Good Health and Well-being (Target 3.6, road safety)

Maintenance reminders and early diagnostic guidance help drivers address vehicle problems before they become unsafe. Roadside assistance gives faster help to drivers in breakdown situations.

SDG 8: Decent Work and Economic Growth (Targets 8.2 and 8.3)

Tsl7tk supports small and medium enterprises such as independent mechanics, workshops, and parts sellers by giving them digital tools and access to more customers. This encourages the formalization and growth of local businesses and creates opportunities in delivery and roadside services.

SDG 9: Industry, Innovation and Infrastructure (Target 9.c)

The platform applies modern technologies, including mobile apps, geolocation, cloud services, an AI-powered chatbot, and a decision support system, to a traditional sector. It also promotes digital innovation among local providers.

SDG 11: Sustainable Cities and Communities

With Cairo as the pilot city, the platform improves access to urban mobility-related services. Better-maintained vehicles and faster breakdown response help reduce road disruption and congestion.

SDG 12: Responsible Consumption and Production (Target 12.5)

Listings for used parts alongside new ones encourage reuse and reduce waste. Transparent stock and availability information also reduces unnecessary trips and wastes time.

SDG 13: Climate Action

Fuel logs with calculated consumption, together with regular maintenance reminders, help drivers spot inefficiency and keep vehicles in good condition. This supports lower fuel use and emissions.

3. Related Work

This chapter reviews existing digital platforms in the Egyptian and regional automotive services market. Each platform is analyzed in terms of its core functionality and, immediately afterwards, its limitations relative to the scope of TSL7TK. The analysis draws on the platforms' official websites, application-store listings, and the literature review in the project proposal. Where a feature is not documented in these sources, it is described as "not documented" rather than as absent, since the lack of public documentation does not prove that a feature does not exist.

3.1 Overview and Gap Analysis of Existing Solutions

1.Atlobha

Atlobha is a spare-parts platform available as a mobile application and an online store. It lets customers request quotations for genuine and aftermarket parts supplied through authorized dealers. It also offers accessories, oils, and tires, mobile services delivered to the customer's location, services at accredited centers, and vehicle transport. Its official sources identify it as operating in Saudi Arabia rather than Egypt, so it is best classified as a regional rather than a local competitor.

Its limitations relative to TSL7TK are as follows. The platform is built around dealer-supplied parts and does not provide a discovery mechanism for independent mechanics or small repair centers. It does not allow customers to compare competing offers from multiple independent sellers. No diagnostic support is documented that helps a driver identify a fault before selecting a service. It also does not serve the Egyptian market, so it cannot address local availability, pricing, or provider networks.

2. 3arabawy

3arabawy is a general automotive marketplace for Android and iOS. It supports buying and selling new and used cars and motorcycles, searching for original and aftermarket parts, and locating maintenance providers. Users can filter listings by make, model, price, and location, and communicate directly with sellers and service providers. Its Google Play listing shows more than 50,000 downloads, which indicates a considerable user base.

Because the platform covers vehicle trading, rentals, parts, and services, its maintenance functions are a secondary component rather than the focus of the user experience. Its search follows a classified-advertisement model, which leads to inconsistent listing quality and gives users no structured way to verify providers. No diagnostic mechanism is documented. It also does not appear to offer an end-to-end repair-request workflow, and it does not link parts to a specific vehicle problem. Maintenance reminders, fuel tracking, and roadside assistance are likewise not documented.

3. EgyCarParts

EgyCarParts is an Egyptian e-commerce website specialized in vehicle spare parts, with a warehouse in New Cairo. The proposal records a catalogue of more than 9,000 parts searchable by make, model, and year. The site offers category browsing, a shopping cart, product comparison, requests for parts that are not in the catalogue, imported products, and a published warranty of up to twelve months on eligible items. Delivery is provided according to its published shipping policy.

Its limitations are as follows. The platform is a retail store and not a marketplace connecting users with independent service providers. Delivery is limited to Cairo under the published policy. The proposal notes that it sells new parts only, so it does not support the reuse of used parts. No mechanic finder, repair booking, diagnostic tool, roadside assistance, or fuel log is documented. A customer must already know which part is needed, because the platform offers no help in identifying the underlying problem.

4. PartFinderEgypt

The proposal describes PartFinderEgypt as a website where customers submit requests for used parts and receive quotations from sellers across Egyptian governorates. The proposal also records that it offers no checkout, is available only as a website, and covers used parts only.

An independent search did not find an official source confirming a platform of this name operating in Egypt. The closest match is a mobile application called Partfinder, which presents itself as a UAE-based marketplace and supports new and used parts, original and aftermarket components, quotations from multiple suppliers, seller verification, and in-app communication. Its store page shows a modest rating of 2.6 out of 5, and some user reviews mention the lack of Arabic-language support and usability difficulties.

The identity and Egyptian presence of PartFinderEgypt therefore remain unverified, and the following should be read as limitations of the model described in the proposal and of the similar UAE application. Neither covers mechanics or repair centers. Neither documents repair booking, diagnostic support, or maintenance and fuel tracking. The multi-quote model addresses only parts sourcing, and its value depends on the number of active sellers.

5. N Auto Express

N Auto Express is an Egyptian automotive company, part of the Nour ElDin ElSherif Group, that operates an online store for spare parts and tires and offers automotive services. Its documented features include a large catalogue (more than 5,300 results in the parts section), original and aftermarket parts, and search by part number or Vehicle Identification Number (VIN) to match compatible parts. It also provides a spare-part request form, a tiered B2B program with volume-based discounts and quotations, and delivery with shipment tracking in Greater Cairo and Alexandria. A maintenance booking page also appears in the site structure.

Its limitations are as follows. The platform is operated by a single company and is not an open marketplace that brings together independent mechanics and sellers. Its fixed pricing offers no competitive quotations. The scope of the maintenance service and its booking process are not described in enough detail to establish an integrated repair workflow. No diagnostic tool is documented that helps drivers understand a fault, and no unified tracking of repair requests between customers and independent mechanics is documented. Fuel tracking and roadside assistance are also not documented.

6. Dawar

Dawar is a mobile application for finding nearby spare-parts shops and maintenance workshops. Users can add their vehicles, filter results by part compatibility, view shops and workshops on a map, save favorites, reserve parts for pickup, and rate providers. Merchants and workshops can manage their products and incoming orders through dedicated tools. The platform describes its role as connecting customers with shops, with agreement and payment taking place directly between the parties and no online payment processing.

Its main limitation is its emphasis on discovery and direct contact, which leaves the customer without a structured path from a vehicle problem to a service and a required part. The proposal records that it has no breakdown → service → provider workflow. The documentation does not show a formal mechanism for confirming and tracking the execution of maintenance work. No diagnostic tool, fuel tracking, or chatbot support is documented.

7. Engazny

Engazny is an Android application that uses the device's GPS to help users find the nearest mechanics, maintenance centers, and spare-parts stores.

Publicly available information about the application is limited, and the following limitations should be read in that light. It is primarily a discovery tool. According to the proposal, it offers no provider status or request tracking. No review and rating system, request confirmation, or provider verification is documented. Maintenance planning, diagnostic support, fuel tracking, and roadside assistance are also not documented.

8. Sayyn

Sayyn is the platform recorded in the proposal under this name; the name "Sayyarat" used in the project brief does not match any source reviewed, and the two are assumed here to refer to the same platform. Sayyn is a car-maintenance application for Android and iOS, together with a companion application for tow-truck drivers. Its official website reports a network of more than 600 service centers and more than 250 tow-truck drivers in Cairo and Giza. It covers periodic maintenance, oil and battery changes, brakes and tires, air conditioning, electrical and engine work, and cleaning. Users can book centers, compare service information and prices, view ratings, track order status, request towing or roadside assistance, and keep a digital maintenance record. Services for corporate fleets are also offered.

Its limitations are as follows. The platform concentrates on service delivery and does not document an integrated marketplace in which sellers compete with multiple quotations, nor a mechanism for verifying that a part is compatible with a specific vehicle. No diagnostic decision-support tool is documented that helps users understand a fault before choosing a service. The publicly stated coverage is limited to Cairo and Giza. The proposal also notes that availability is not alive, although this cannot be confirmed without direct testing of the application.

9. Ringaat

Ringaat is a marketplace connecting vehicle owners with parts merchants and workshops, available on Android, iOS, and the web. Customers can post a part request with a photo or voice message and receive quotations and photos from merchants, who compete to offer the best terms. The platform supports comparison of price, part condition, and merchant specialization. It provides a virtual garage, entry of vehicle details or a VIN, search for mechanics and parts shops, mileage-based maintenance reminders, indicative fair-price trends for certain parts, and merchant ratings and locations. Communication with sellers is largely handled through WhatsApp.

Its limitations are as follows. The platform's evident focus is parts sourcing, and it does not offer a repair journey managed inside the application. The proposal notes that it is not built around a problem-to-provider flow. Purchases and follow-up depend on direct contact with the merchant. No fault diagnosis through a decision-support system, fuel-consumption tracking, or integrated roadside assistance is documented. Public user reviews on application stores were too limited to draw conclusions about service quality.

10. Dynamu

Dynamu is an Egyptian automotive marketplace available as a website and as iOS and Android applications, connecting customers and workshops with authorized spare-parts retailers. Its website states a catalogue of more than 300,000 parts and accessories, searchable by vehicle make, model, year, and engine type. Orders are delivered to the customer's address across all 27 governorates, with payment by cash on delivery, cards, or digital wallets and a seven-day return policy for unused parts. Workshops can place bulk orders, and retailers manage their own inventory through a dashboard. The website also states that it can refer customers to mechanics for installation. Its limitations are as follows. The platform is centered on parts sourcing and delivery. The referral to mechanics is only a service the company can arrange, and no mechanic search, repair booking, or request tracking is documented. A customer must still know which part is needed, because no diagnostic support is documented. No maintenance reminders, fuel log, roadside assistance, or competing-quotation model is described. The figures above come from the company's own website and have not been independently verified.

3.2 Identified Market Gaps and How TSL7TK Differs

The platforms above fall into three groups. The first is parts retailers and marketplaces (Atlobha, EgyCarParts, PartFinderEgypt, N Auto Express, Ringaat), which help customers find and buy parts but do not support the repair itself. The second is service and discovery platforms (Sayyn, Dawar, Engazny), which help customers find providers but typically offer little help with parts sourcing, or none. The third is general marketplaces (3arabawy), whose maintenance functions form only a small part of a broader vehicle-trading offering. Four recurring gaps emerge from this analysis. TSL7TK addresses each of them as follows.

Gap 1: The absence of an integrated problem-to-repair journey. Existing platforms handle either parts or services, and customers must move between several applications to complete one repair. TSL7TK links a vehicle problem to a suitable provider and to the required part within a single flow, covering location-based search, provider information, and service or parts requests.

Gap 2: The lack of diagnostic support. None of the reviewed platforms documents a diagnostic decision-support system. Drivers who cannot name a faulty part or describe the issue are left to guess. TSL7TK integrates a diagnostic decision-support system into an AI-powered chatbot. This system helps drivers identify likely causes and the type of service required before they contact a provider. Its output is presented as an indicative assessment, not a confirmed diagnosis, and refers the driver to a specialist inspection.

Gap 3: Weak request management and accountability. Most platforms stop at discovery or direct contact, so requests cannot be tracked reliably. TSL7TK lets drivers submit service and parts requests and track provider confirmation, completion, or cancellation, with notifications on status changes. Reviews are linked to completed requests, which gives them a basis in real transactions.

Gap 4: Limited reliability and transparency of listings. Classified-style listings, outdated stock information, and unverified providers reduce user trust. TSL7TK addresses this through administrator verification of provider information, moderated reviews with abuse reporting, and visible update times on prices, availability, and stock. Providers manage their own profiles, listings, and incoming requests, which keeps the platform from holding inventory or operating workshops.

Gap 5: Fragmented ownership and maintenance support. Maintenance reminders and fuel tracking are either absent or only partly present in the reviewed platforms. TSL7TK stores vehicle details in a user garage, records maintenance dates, mileage, and fuel purchases, and calculates consumption once enough entries exist.

Gap 6: Disconnected emergency and delivery services. Roadside assistance and parts delivery are provided by separate services, if at all. TSL7TK includes location-based roadside service requests and parts delivery in the same application, so that emergency, repair, and sourcing needs are handled together.

Gap 7: Limited geographic and market fit. Some platforms serve other countries (Atlobha, Partfinder), and others are limited to specific cities (Sayyn, EgyCarParts, N Auto Express). TSL7TK is designed for the Egyptian market, with a first release piloted in Cairo, which allows provider networks and data to be validated before wider expansion.

Taken together, TSL7TK does not seek to replace any single existing platform in the individual function it performs best, such as catalogue depth or tow-truck network size. Its contribution is the integration of diagnosis, provider selection, parts sourcing, request tracking, maintenance planning, and emergency support into one coherent experience. As noted earlier, the absence of a documented feature in a competing platform is not conclusive evidence that it does not exist, and these gaps should be validated through direct testing of the applications.

4. Problem Statement:

Car owners in Egypt face several challenges when their vehicles require maintenance or repair, particularly when searching for suitable mechanics, repair centers, and spare-parts sellers. Finding a specific spare part can be a time-consuming process, as customers often have to ask multiple sellers and search in different places until they find the required part. In some cases, customers may also struggle to determine whether the available spare parts are original or reliable, due to limited information about their quality and authenticity.

Another major challenge occurs when a vehicle breaks down unexpectedly. Car owners may find it difficult to locate nearby mechanics or repair centers, reach their locations, or identify where to go and whom to contact for assistance. This becomes even more challenging when users need urgent support and do not have sufficient information about available service providers in their surrounding areas.

Although several automotive platforms exist in the Egyptian market, most focus on specific services, such as selling spare parts, displaying products, or helping users locate mechanics. However, there is still a gap in providing a connected and reliable experience that brings together spare-parts sellers, mechanics, and repair centers while helping car owners find suitable services and required parts based on their location and needs. The lack of integration between these services, combined with difficulties in accessing reliable information about providers, spare-parts authenticity, availability, and nearby services, leaves car owners dependent on repeated searching and asking different people to find appropriate automotive solutions.

5. Objectives:

5.1 General Objectives:

The general objective of the Tsl7tk App is to develop an integrated mobile Application platform that connects car owners in Egypt with mechanics, repair centers, and spare-parts sellers. The platform aims to simplify access to automotive services and required parts, improve communication between customers and service providers, and support better decision-making through integrated diagnostic assistance and vehicle maintenance management features.

5.2 Specific Objectives:

The project aims to achieve the following specific objectives:

1) Help Drivers Locate Suitable Local Services and Parts Sellers:

Provide car owners with a wide range of spare parts and automotive products that match their vehicle types and needs, while enabling them to locate nearby mechanics, repair centers, and spare-parts sellers based on their location and required services.

2) Link Service Requests to Provider Confirmation and Relevant Parts:

Allow customers to submit service or spare-parts requests, receive provider confirmations, and track request status.

3) Improve Listing Reliability Through Provider Checks and Moderated Reviews: Ensure the credibility of information displayed on the platform by verifying provider identities and details, reviewing submitted information, keeping listings updated, and moderating customer reviews to maintain accurate and trustworthy content.

4) Support Maintenance Planning and Fuel-Consumption Tracking:

Provide features that help car owners record maintenance dates, vehicle mileage, and fuel purchases to support better vehicle management. To help car owners maintain their vehicles through regular and well-organized maintenance schedules.

5) Let Providers Manage Their Profiles, Listings, and Incoming Requests:

Allow service providers and spare-parts sellers to manage their business information, services, availability, prices, stock status, and customer requests.

6) Support Customers Through a Chatbot with an Integrated Car Diagnostic

DSS: Provide an intelligent chatbot integrated with a Decision Support System (DSS) to help customers understand possible vehicle problems and receive diagnostic assistance.

7) Provide Spare-Parts Delivery and Roadside Assistance Services:

Enable customers to request spare-parts delivery directly to their preferred location, such as their home, and request roadside assistance when their vehicles break down, allowing service providers to reach them based on their current location and needs.

6. Significance:

The Tsl7tk project is important because car owners in Egypt often face difficulties when searching for suitable spare parts, trusted mechanics, and nearby repair centers. Finding the required parts or services can take time and effort, especially when customers have limited information about product availability, spare-parts reliability, and service providers. Therefore, the project aims to make automotive services easier to access, improve the connection between customers and service providers, and provide a more organized and reliable experience. The project benefits several groups:

6.1. Car Owners:

Car owners benefit from easier access to suitable spare parts, mechanics, and repair centers. This helps them save time and effort, find services that match their needs and vehicle types, and receive support when facing unexpected vehicle problems.

6.2. Mechanics and Repair Centers:

Mechanics and repair centers benefit from reaching more potential customers and increasing the visibility of their services, helping them connect with customers who need maintenance and repair services.

6.3. Spare-Parts Sellers:

Spare-parts sellers benefit from having a platform to display their products and reach customers by searching for specific spare parts, creating more opportunities to receive customer requests and increase their sales.

6.4. Automotive Service Market:

The automotive service market benefits from improved communication and connections between customers and service providers, greater access to automotive information, and more organized coordination of services

7 System Request

7.1 Project Sponsor

The TSL7TK project team (sponsor to be confirmed).

7.2 Business Need

This project has been initiated to solve the difficulties that car owners in Egypt face when they need spare parts and automotive services. Today, a driver who needs a part or a repair has to search through several sources and contact many businesses one by one, with no single place to see what is available, at what price, and from whom. Drivers currently face the following problems:

- Finding a specific spare part takes a great deal of time and effort, as drivers travel between shops and make many calls.
- Drivers who cannot go out to look for a part, or do not have time in their schedule to visit shops, have no convenient way to get the part delivered to them.
- Drivers cannot easily compare prices with sellers, so they may be overcharged or misled about the price or the condition of a part.
- Drivers often do not know where to go or whom to trust and may end up visiting unreliable shops.
- There is no easy way to know whether a mechanic's or seller's information and prices are current, or to see how other drivers rated them.
- Drivers find it hard to keep up with vehicle maintenance and to track their fuel use.
- When a car breaks down, drivers struggle to understand the problem and to get help quickly.

TSL7TK is needed to give drivers a reliable place to find trusted mechanics and spare-parts sellers, compare their options, have parts delivered when they cannot go out, and manage their vehicles with less effort.

7.3 Business Requirements

Using the mobile application, drivers will be able to find mechanics near them and reach them directly through map directions, look up spare parts and see the nearest sellers that list them, call for emergency roadside assistance, and request delivery of spare parts to their location. Providers will use the same application to manage their own profiles, services, and spare-parts listings and to receive and respond to parts requests. The specific functionality that the system should have includes the following:

- Search for mechanics and spare-parts sellers by location and category.
- View provider profiles with services, working hours, and ratings.
- Contact providers and use map directions to reach them.
- Publish and update spare-parts listings with part numbers, prices, and last update time.
- Request delivery of spare parts to the driver's location. The request goes to the provider, who accepts or declines it, and it remains pending until the provider decides.
- Track each part's request as pending, accepted, or declined, with notifications on status changes.
- Review providers after completing requests, with abuse reporting, moderation, and provider verification by administrators.
- Store vehicles in a personal garage, record maintenance with reminders, and log fuel and mileage to calculate consumption.
- Get help from an AI-powered chatbot with a Diagnostic Decision Support System to understand potential vehicle problems.
- Request roadside assistance through an emergency button that sends a towing vehicle to the driver's location when the car breaks down.
- Provide role-based access for drivers, providers, and administrators.

7.4 Business Value

We expect that TSL7TK will save drivers money and time by helping them compare nearby mechanics and spare-parts sellers, keep track of vehicle maintenance, and monitor their fuel use. We expect providers to gain new customers and reduce their marketing costs by reaching drivers through one application. We also expect both groups to save time by handling searches, requests, and listings in one place.

Conservative estimates of the yearly value to each driver include the following:

- EGP 1,200 in savings from comparing spare-parts prices.
- EGP 1,500 in savings from avoided repairs through maintenance reminders.
- EGP 3,000 in savings from monitoring fuel use.
- EGP 600 in savings from fewer unnecessary trips.

These estimates total EGP 6,300 per driver per year, in addition to about 20 hours of time saved.

Conservative estimates of the yearly value to each provider include the following:

- EGP 36,000 in additional revenue from new customers reached through the app.
- EGP 6,000 in marketing cost savings.

These estimates total EGP 42,000 per provider per year, in addition to about 10 hours of time saved each month. **All estimates will be refined with the supervisor and market research.**

7.5 Special Issues or Constraints

- TSL7TK is a connector platform: it does not own or operate workshops and does not hold spare-parts inventory. Providers are responsible for keeping their own prices and parts listings up to date, so every listing displays its last update time and demo records are clearly labelled. Stock levels are not tracked within the app. Instead, a provider accepts or declines each parts request, and a request that is declined or not accepted generally indicates that the part is unavailable.
- Many mechanics have limited familiarity with technology, so persuading most of them to adopt the app may be a significant challenge. The app must therefore be simple for providers to use, and they may need guidance during onboarding.
- In its initial stage, the project is constrained to Cairo only, including the roadside assistance service. If the platform proves successful, coverage will be expanded to other governorates.
- The project depends heavily on external businesses, so TSL7TK has limited control over the quality of the services that customers receive. For example, roadside assistance will require a contractual agreement with a towing company, and parts delivery will be carried out by a third-party delivery company. Any failure by either company, such as delays in reaching the customer, would be reflected on the platform. Similarly, providers may misrepresent the condition of a part, for instance by describing a used part as new.

8. System Scope

8.1 System Overview

TSL7TK is a mobile-based automotive utility platform that connects car owners in Egypt with mechanics and spare-parts sellers. In this document, **mechanics** refers to repair centers and workshops, and a business may offer repair services, sell spare parts, or do both. The platform gives drivers one place to discover mechanics, search for spare parts, and request roadside assistance and parts delivery, while giving providers one place to manage their business information and incoming parts requests.

Beyond discovery, the system offers vehicle management (garage, maintenance reminders, fuel logging), AI-powered chatbot support integrated with a Diagnostic DSS, and requests for roadside assistance and spare-parts delivery. The platform starts as a pilot in Cairo and may expand in later phases

8.2 In-Scope Features

Module	Description
User Management	Account registration, login, and role-based access control for drivers, providers, and administrators. This module determines who may sign in and which functions each role may use.
Vehicle Garage	Drivers store their vehicles using make, model, and year; this data supports maintenance records and fuel tracking.
Provider Management	Providers manage business profiles, locations, contact details, services, and working hours.
Spare-Parts Management	Sellers publish parts with part number, price, and last update time.
Search and Filtering	Location-based search for mechanics and spare-parts sellers, filtered by distance and category.

Module	Description
Provider Profiles	Drivers view mechanic and seller profiles showing services, working hours, and ratings.
Parts Locator	Drivers search for a specific part and see the nearest sellers that list it, then request delivery of the part.
Reviews and Ratings	Reviews allowed only for completed requests, with abuse reporting and moderation.
Notifications	Users are notified when the status of a parts request change.
Contact and Directions	Contact actions and map directions that lead drivers directly to mechanics and spare-parts sellers.
Maintenance Management	Maintenance dates, records, and reminders.
Fuel Tracking	Mileage and fuel purchase logging; consumption calculated once enough entries exist.
AI Chatbot and Diagnostic DSS	Customer support and assistance in assessing potential vehicle problems.
Roadside Assistance	Emergency button that sends a request for a towing vehicle to the driver's location when the car breaks down.
Parts Delivery	Customers request delivery of spare parts to their location. The provider accepts or declines the request, which stays pending until decided. A third-party company carries out the delivery.
Administration	Administrators verify providers' business information, manage disputed or outdated listings, and moderate reviews and abuse reports. This module concerns the trustworthiness of provider information, not account access.

Table 8.1: Functional Features

8.3 Out-of-Scope Items

- **Workshop ownership and operation:** TSL7TK will not own, manage, or operate repair workshops.
- **Inventory ownership:** the platform will not purchase, store, or maintain spare-parts inventory.
- **Direct repair services:** repairs are performed by independent providers, not by the platform.
- **Manufacturing Parts:** the platform will not manufacture automotive components.
- **Guaranteed real-time inventory:** stock status is not tracked in the app; providers accept or decline each parts request.
- **Nationwide initial launch**: the pilot is limited to Cairo.
- **Delivery operations:** spare parts delivery is carried out by an independent third-party delivery company, not by platform.

8.4 System Users and Responsibilities

Role	Main capabilities
Drivers (Customers)	Manage account and garage; search mechanics and parts; request roadside assistance; request spare-parts delivery and track the parts request status; view profiles and ratings; review completed requests and report abuse; record maintenance and fuel; use chatbot and DSS.
Providers	Manage business profile, locations, and contact details; publish services and parts; update prices and working hours; accept or decline incoming parts requests; access AI-powered chatbot support.
Administrators	Verify provider business information; manage disputed and outdated listings; moderate reviews and handle abuse reports; maintain information reliability and any administrator responsibility.

Table 8.2: User Roles and Capabilities

8.5 System Boundaries

The system begins when a user opens the mobile application and interacts with its features. It includes account management, vehicle information, provider and parts listings, search, requests, reviews, notifications, maintenance records, fuel tracking, chatbot support, diagnostic DSS, roadside assistance, and parts requests.

The system ends at facilitating connections and requests between customers and independent providers. Repair work, ownership of parts inventory, parts delivery, and roadside towing remain the responsibility of independent providers and third-party companies.

8.6 Assumptions

- Providers will register directly and keep their own information current.
- Cairo has enough mechanics and parts sellers willing to join for a meaningful pilot.
- Drivers have smartphones with location services and internet access.
- Independent third-party delivery and roadside companies will be available to fulfil those requests.
- The selected map and data services fit within the project's cost limits.

8.7 Success Criteria

- Drivers can complete realistic tasks such as finding a mechanic, finding a part, requesting
- roadside assistance or parts delivery, and logging fuel.
- Providers can complete realistic tasks such as updating their services and parts listings and handling a parts request.
- Reviews are only possible after completed requests.
- Listings show last update times and demo records are clearly labelled.
- Each role can access only the functions allowed for it.
- Usability findings from drivers and providers are collected and used to improve the prototype.

9. SWOT Analysis

A SWOT analysis identifies the internal strengths and weaknesses of the project and the external opportunities and threats it faces. It is based on the system scope and on the market, technical, operational, economic and organizational analysis in Section 10. The results are summarized in Table 9.1 and Table 9.2.

Strengths	Weaknesses
• One platform combining service providers, parts sellers, requests, a maintenance and fuel log, a diagnostic chatbot, and delivery and roadside services • Trust mechanisms: provider verification, reviews linked to completed requests, abuse reporting and moderation • Three-role design (driver, provider, administrator) that serves both sides of the market • Low-cost, scalable stack built on PostgreSQL with PostGIS, with local payment gateways and strong Egyptian map coverage • Focused pilot in one city (Cairo)	• Student team of five with a limited budget and no existing provider network • Wide scope (eight modules) relative to team size • Dependence on providers to keep availability, prices and stock current • Diagnostic knowledge base requires mechanic input to be reliable • Vehicle reference data is United States oriented and provider listings are sparse at launch

Table 9.1: Internal analysis of Tsl7tk

Opportunities	Threats
• Large and growing vehicle fleet, with Cairo the largest single market (Section 10.2) • High internet and mobile adoption, which supports a mobile-first service • Gaps left by existing platforms (Section 3) • Several revenue paths: boosting, subscriptions, service commissions and advertising • Partnerships with parts sellers and roadside providers	• Free informal alternatives: social-media groups and general map applications • Existing platforms that may add similar features • Fake or abusive reviews that weaken trust • Providers who bypass the platform to avoid fees • Data-protection obligations and changes in cloud, map or AI service prices and exchange rates

Table 9.2: External analysis of Tsl7tk

The analysis shows that the project's main **advantage** lies in combining services, trust and vehicle management in one application, while its main **weaknesses** are capacity and data dependence. Success therefore depends on building a reliable provider base and keeping the scope manageable.

10 Feasibility Study

10.1 Overview and Approach

A feasibility study determines whether a proposed system can be developed and operated successfully within realistic constraints. This section evaluates Tsl7tk across five dimensions: **market**, **technical**, **operational**, **economic** and **organizational** feasibility. Each dimension is judged against the criteria in Table 10.1, and the section closes with a risk assessment and an overall conclusion. Project scheduling is not assessed here; it is covered in the Work Plan (Section 13).

Dimension	Key question	Evidence used
Market	Is there enough demand, and is there a gap that existing solutions leave open?	Official vehicle statistics, digital-adoption data, related work, user survey
Technical	Can the system be built with the chosen technology and available data?	Technology evaluation, data-source review, prototype tests
Operational	Will drivers, providers and administrators actually use and sustain the system?	Role analysis, workflow design, provider-onboarding plan
Economic	Do expected revenues justify the costs?	Cost estimate, revenue model, break-even analysis
Organizational	Does the team have the structure, commitment and resources to deliver and sustain the project?	Responsibility allocation, capacity review, stakeholder commitments, continuity plan

Table 10.1: Feasibility dimensions and assessment criteria

10.2 Market Feasibility

Market size and growth. The pilot city's vehicle population is large and growing. The Central Agency for Public Mobilization and Statistics (CAPMAS) reported that licensed vehicles in Egypt rose 6% to 11.05 million in 2025, with Cairo holding the largest share at 2.84 million vehicles, about 25.7% of the national total [1]. Passenger cars made up 6.0 million vehicles, or 54.1% of the fleet [1]. For comparison, the fleet stood at 9.95 million licensed vehicles at the end of 2023 [2]. That is growth of roughly 11% in two years, which means a growing base of vehicles needing maintenance, parts and repair.

Indicator	Value	Source
Licensed vehicles, Egypt (end of 2023)	9.95 million	[2]
Licensed vehicles, Egypt (2025)	11.05 million	[1]
Licensed vehicles, Cairo (2025)	2.84 million (25.7% of national)	[1]
Passenger cars, Egypt (2025)	6.0 million (54.1% of fleet)	[1]

Table 10.2: Licensed vehicles in Egypt and Cairo

Digital readiness. A mobile-first platform needs users who are already connected. Egypt had 98.2 million internet users at the end of 2025, equal to 82.7% of the population [3], and mobile internet users reached 92.6 million in December 2025 [4]. Smartphone-based discovery and requests are therefore a realistic channel for the target segment.

Target segments. The platform serves two groups:

- **Drivers:** private car owners in Cairo who need maintenance, repair or parts.
- **Providers:** independent mechanics, repair centers and spare-parts sellers operating in Cairo.

Market sizing. Table 10.3 follows the standard total, serviceable and obtainable market approach. The first two figures come from official statistics. The third is a planning assumption, not a forecast.

Level	Definition	Value
Total market	Licensed vehicles in Egypt	11.05 million [1]
Serviceable market	Licensed vehicles in Cairo (pilot city)	2.84 million [1]
Obtainable market (assumption)	0.5% of Cairo vehicles reached during the pilot	About 14,200 vehicles

Table 10.3: Market sizing for the Cairo pilot

The supply side (workshops and parts shops in Cairo) will be estimated by counting relevant business categories through Google Maps Platform place searches in Cairo, then adjusted through direct field registration. The result will be inserted here once the count is complete: [number of providers].

Competitive landscape. Existing car service and spare-parts platforms are reviewed in Section 1.3 (Related Work), together with the gaps they leave open. For market feasibility, the relevant point is that entry depends on differentiation. The intended differentiator of Tsl7tk is a single platform that combines service providers, parts sellers, request-linked and moderated reviews, vehicle maintenance and fuel records, a diagnostic chatbot, and delivery and roadside services. Indirect alternatives also compete for the same drivers: general-purpose map applications, social-media groups and personal recommendations. These cost nothing to use, so the platform must give drivers clearly better reliability and convenience to win them over.

Market feasibility rating: feasible with conditions. The fleet is large and growing, and connectivity is high. The rating is conditional on the survey confirming demand and on the gaps identified in Related Work being real.

10.3 Technical Feasibility

Technical feasibility asks whether Tsl7tk can be built with the chosen technology, the available data and a team of five. The assessment covers the architecture, the technology stack, the data sources, the two advanced modules (diagnostic chatbot, and delivery and roadside services) and the main technical constraints.

Proposed architecture. The system has three layers, shown in Figure 10.1. The client layer is a Flutter mobile application for drivers and providers and a web dashboard for administrators. The service layer, built on Supabase or a custom NestJS or FastAPI backend, provides phone-number authentication, a PostgreSQL database with PostGIS, search and server-side logic. The data and integration layer connect map services, push notifications, payment gateways, the diagnostic chatbot, courier and two partners, and vehicle reference data.

Technology stack. The mobile interface will be built with Flutter and Dart, supporting Android and iOS, Arabic and English, and right-to-left layout, with Android released first. Data will be held in PostgreSQL with PostGIS for distance-based provider search, behind a backend built on Supabase or a custom NestJS or FastAPI service. Authentication uses phone-number OTP, payments use local Egyptian gateways, and a web dashboard supports provider verification and moderation. The diagnostic chatbot uses a language model for conversation and a rule-based decision tree for diagnosis. Delivery and roadside services rely on third-party couriers and two partners instead of a platform-owned fleet. Billable services such as SMS, payments and AI calls must be budgeted separately, as assessed in Section 10.5.

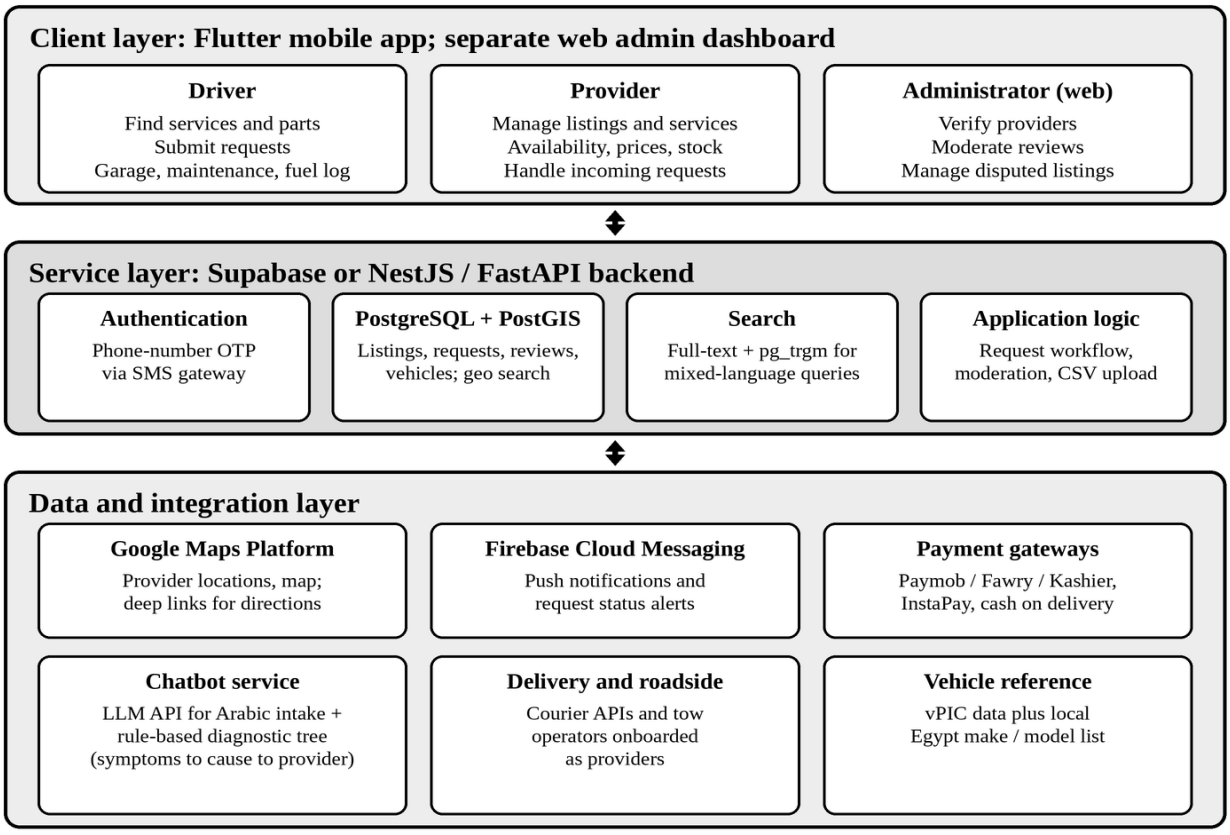


Figure 10.1: Proposed system architecture

Component	Technology	Purpose
Mobile interface	Flutter, Dart	One codebase for Android and iOS; good Arabic and right-to-left support; Android released first
Database	PostgreSQL with PostGIS	Relational data (vehicle, part compatibility, provider, request, review); native nearest-provider search within a set distance

Component	Technology	Purpose
Backend	Supabase, or NestJS / FastAPI	Supabase provides database, authentication, real-time and storage out of the box; a custom API gives more control and system-design depth
Search	PostgreSQL full-text search with pg_trgm; Meilisearch or Typesense later	Typo-tolerant search for part numbers and mixed Arabic, English and Franco-Arabic queries
Maps and location	Google Maps Platform	Best Egyptian place and Arabic address coverage; deep links for directions to avoid Directions API costs
Notifications	Firebase Cloud Messaging	Free push notifications, independent of the backend
Authentication	Phone-number OTP through an SMS gateway reliable in Egypt	Matches how Egyptian users, garages and parts sellers sign in
Payments	Paymob, Fawry or Kashier, plus InstaPay and cash on delivery	Subscriptions, boosted listings and delivery fees through local gateways
Diagnostic support	LLM API for intake with a rule-based diagnostic decision tree	The LLM handles Egyptian-Arabic conversation and symptom extraction; a deterministic tree maps symptoms to likely causes, service category and provider search

Component	Technology	Purpose
Administration	Web dashboard (Next.js, or Appsmith / Retool)	Provider verification and moderation are impractical on a phone
Delivery and roadside	Third-party couriers (e.g., Bosta, Mylerz) and tow providers	Tow operators are onboarded as providers and couriers are used through APIs, so the platform runs no fleet or inventory; API fit to be confirmed
Design, version control and testing	Notion, Figma, GitHub with Actions CI, Postman	Planning, wireframes, code management, automated builds and interface testing

Table 10.4: Proposed technology stack

This stack is a proposal that will be reviewed against the final requirements before adoption.

Data sources. Provider and location data will come from direct provider registration, supported by bulk CSV or Excel upload of parts and manual onboarding by the project team, because small Egyptian sellers keep their stock in spreadsheets and messaging apps. Place and address data will use Google Maps Platform, which offers the strongest Egyptian business and Arabic address coverage among the options reviewed. Listings will not be scraped from competing platforms, because those risks breaching their terms, the data goes stale, and providers are expected to update their own stock. Any demonstration records will be clearly identified. Vehicle reference data will use the NHTSA vPIC catalogue, which decodes vehicle identification numbers, offers an API with JSON, CSV and XML output, and is intended for model years 1981 onward [7]. Because vPIC is built on United States regulatory data, it may not cover every model sold in Egypt, so a locally checked reference list of makes and models will be kept alongside it.

Diagnostic chatbot and DSS. The chatbot combines two parts. A knowledge-based decision support system maps reported symptoms to probable causes, urgency and recommended actions through explicit rules. A large language model (LLM), accessed through an API, understands free-text questions in Arabic and English and guides the driver through follow-up questions. The LLM only handles the conversation and extracts symptoms; the diagnosis itself comes from the rule-based decision tree (symptoms, likely causes, service category, provider search), because an LLM-only diagnosis is a liability risk and hard to defend. The rules keep diagnostic output consistent and checkable, while the language component keeps the conversation natural. The main dependency is the knowledge base. It will be built from mechanic interviews and technical documentation, starting with a limited set of common fault categories [list of fault categories], and each rule will be validated by a practicing mechanic before release. Output is advisory and always recommends confirmation by a provider.

Delivery and roadside services. At the system level, these services need request creation with a location pin, assignment to a provider or partner, status tracking and notifications. These functions reuse the request workflow that already exists for service and parts requests, so the technical risk is low. The platform will not operate vehicles or workshops itself: tow operators are onboarded as providers and deliveries use courier APIs such as Bosta or Mylerz, whose fit must be checked before commitment. The first release follows a simple flow (request, nearest providers notified, then a call or WhatsApp contact), and live tracking is deferred because it needs background services and realtime infrastructure. The physical side depends on partners and is assessed under operational feasibility.

Technical constraints and responses. Table 10.5 lists the main constraints and the planned response to each.

Constraint	Impact	Planned response
Mixed Arabic, English and Franco-Arabic part queries	Missed or wrong search results	Typo-tolerant search with pg_trgm; Meilisearch or Typesense later
Sparse provider data at launch (cold start)	Empty marketplace on day one	Bulk CSV / Excel upload, manual onboarding, direct registration and admin verification
vPIC covers mainly United States models	Missing or wrong vehicle details	Local, validated make and model reference list
Map, SMS, payment and LLM usage costs	Budget overrun	Usage budgets and alerts, caching, deep links instead of Directions API calls
DSS knowledge-base accuracy	Incorrect advice	Limited validated fault categories and advisory wording
Variable mobile connectivity	Failed requests	Offline data caching and retry logic

Table 10.5: Technical constraints and planned responses

Team capability and testing. The team of five will divide responsibilities across mobile development, back end and data, DSS knowledge base, interface design and testing. Delivery will be tireded: an MVP (accounts, garage, search, requests, reviews and administration), then the chatbot and DSS, then roadside and delivery as partner integrations. Current skills in Dart, Flutter, PostgreSQL and the chosen backend: [team skills to be confirmed]. Testing will use emulators and Postman for interfaces and services, followed by usability tests in which drivers and providers complete realistic tasks.

Technical feasibility rating: feasible with conditions. The core technologies are mature and low-cost at prototype scale. The rating depends on three conditions: map, SMS and LLM usage costs are kept within budget, the DSS scope stays limited to validated fault categories, and provider data quality is secured through direct registration.

10.4 Operational Feasibility

Operational feasibility asks whether the people who must use and run the system will do so, and whether the processes it depends on can be sustained.

Roles and responsibilities. Drivers use the application to find services and parts, submit requests and keep vehicle records. Providers (mechanics, repair centers and parts sellers) maintain their own profiles, services, availability, prices and stock, and respond to requests. Administrators verify providers, moderate reviews and handle disputed or outdated listings. In the pilot, administration is performed by the project team.

Key operating processes. Table 10.6 sets out who is responsible for each process and how it is controlled.

Process	Responsible	Control
Provider onboarding	Provider, administrator	Direct registration; administrator checks business details before a listing is marked as verified
Listing upkeep	Provider	Providers update availability, prices and stock; listings not updated for a set period are flagged as outdated and reviewed
Request handling	Driver, provider	Request, provider confirmation, then completion or cancellation, with the status visible to both sides

Process	Responsible	Control
Review moderation	Administrator	Reviews allowed only for completed requests; abuse reporting; moderator queue
Dispute handling	Administrator	Decision based on request history; disputed listings can be suspended
Delivery and roadside coordination	Partner provider, administrator	Partner terms of service; status tracking; driver notifications; the platform does not operate vehicles
Chatbot knowledge upkeep	Project team, mechanics	Periodic rule review; driver feedback on answers; mechanic validation of changes

Table 10.6: Operating processes and controls

Adoption and incentives. Providers will adopt the platform if it brings them customers at low cost. The first release therefore offers a free basic listing, direct request leads and optional paid promotion. Drivers will adopt it if the information is reliable and the service is free to use, and the maintenance and fuel log gives them a reason to return. Trust is central: online feedback mechanisms can build trust between strangers but are open to manipulation and to weak incentives to leave honest feedback [8]. The design responds by linking reviews to completed requests and adding abuse reporting and moderation.

Support and workload. Many providers may be unfamiliar with listing tools, so an Arabic onboarding guide and a short in-app walkthrough will be provided, together with a basic support channel. Administrator workload grows with the number of providers. Part-time moderation is enough for the pilot, and automated checks such as update reminders and document uploads will reduce manual work as volume grows. The cost of moderation is included in Section 10.5.

Operational feasibility rating: feasible with conditions. The processes are simple and fit the three roles. The rating depends on enough providers registering and keeping their listings current, and on moderation capacity growing with usage.

10.5 Economic Feasibility

Economic feasibility asks whether the expected revenue justifies the cost of building and running the platform. All monetary amounts are expressed in Egyptian pounds (L.E). Paid pilot figures are planning assumptions, not quotations, and must be updated when real prices and survey results are available. Foreign-priced services retain the original study's conversion assumption of L.E 52 per US dollar; this is not a current exchange-rate quotation. Development labour of the five-member student team is not costed; a commercial launch would require paid development and operations staff.

Graduation prototype. Table 10.7 records the actual tool charges for the selected free configuration. The L.E 0 entries reflect deliberate use of free software and no-cost service tiers, subject to the limits below. They do not imply that the later public pilot has no operating costs.

Tool used	Actual cost	Why it is free and key limits
Flutter and Dart	L.E 0	Open-source SDK and language; no licence fee. Existing computers are used.
Supabase Free plan (assumed prototype backend)	L.E 0	Free PostgreSQL database with PostGIS, authentication and storage within limits (500 MB database, 50,000 monthly active users); free projects pause after a week of inactivity [5].

Tool used	Actual cost	Why it is free and key limits
GitHub Free	L.E 0	Free repositories. Private-repository Actions allowance: 2,000 minutes/month and 500 MB artifact storage; stay within included limits [14].
Figma free tier	L.E 0	Starter supports up to 3 collaborative design files with 3 pages each; paid features are excluded [15].
Google Maps Platform	L.E 0	Free monthly usage per SKU (10,000 events for Essentials SKUs), then pay-per-use. Directions use deep links to avoid paid API calls [6, 16].
Firebase Cloud Messaging	L.E 0	Push notifications are a no-cost Firebase product with no usage quota [13].
Emulators	L.E 0	Local Android emulators and a local PostgreSQL with PostGIS database on existing hardware; no paid cloud-device testing or new equipment.
Postman Free	L.E 0	Individual API testing with core free tools. Shared team workspaces and paid automation are excluded; collections are shared through GitHub [17].
Notion	L.E 0	Free plan used for planning and documentation; plan limits to be confirmed before relying on it.
Admin web dashboard (Next.js, or Appsmith / Retool)	L.E 0	Next.js is open source and can run locally for the demonstration; Appsmith or Retool free tiers, if chosen, have their own limits (to be confirmed). Public hosting would be added to the pilot cloud services budget.
Total direct tool cost	L.E 0	Applies only to this free graduation-prototype configuration.

Table 10.7: Graduation prototype tool costs (L.E)

Free-tier limits. The Supabase Free plan includes two active projects with 500 MB of database storage, 1 GB of file storage, 5 GB of egress and 50,000 monthly active users, and pauses projects after one week of inactivity [5]. Usage must remain within these limits. Google Maps Platform gives a free monthly allowance for each SKU and bills per event beyond it [6, 16], so directions are opened through deep links. Firebase Cloud Messaging is listed as a no-cost product [13].

Scope of the zero. L.E 0 is the direct charge for these tools, assuming existing laptops and internet access. Student labour, existing equipment and connectivity are not valued here. App-store publication, paid AI, map and SMS services, payment-gateway fees, legal set-up, marketing and staffing belong to the separate pilot budget. Paid integrations are mocked or deferred in the free demonstration.

Paid pilot costs. Table 10.8 lists the later operating assumptions. Converted store fees are L.E 5,148 per year for Apple, equivalent to L.E 429 per month, and L.E 1,300 once for Google Play [9]. These fees are not needed for an emulator-based graduation demonstration.

Cost item	Basis	One-off (L.E)	Monthly (L.E)
Cloud services (Supabase or hosted backend)	Assumed L.E 2,600 per month above free quotas	—	2,600
Map and geolocation	Assumed L.E 1,560 per month; Google Maps Platform, with deep links for directions	—	1,560
AI chatbot service	Assumed L.E 2,080 per month	—	2,080

Cost item	Basis	One-off (L.E)	Monthly (L.E)
Apple Developer Program	L.E 5,148 per year, spread monthly	—	429
Google Play registration	L.E 1,300, one time	1,300	—
Moderation and verification	Part-time staff	—	4,000
Marketing and provider onboarding	Field registration and digital promotion	—	5,000
Legal and data-protection set-up	Advice and registration	30,000	—
SMS gateway for OTP login	Per-message charges; provider and volume to be confirmed (must deliver reliably in Egypt)	—	[to be estimated]
Payment-gateway fees	Per-transaction fees on subscriptions, boosts and delivery fees (Paymob, Fawry or Kashier); normally deducted from revenue	—	[per transaction]
Courier and tow partner API costs	Depending on partner terms (e.g., Bosta, Mylerz); API fit and pricing to be confirmed before commitment	—	[to be confirmed]
Contingency (10%)	Allowance for price changes	3,130	1,567
Total		34,430	17,236

Table 10.8: Cost assumptions (L.E)

The three bracketed items (SMS gateway, payment-gateway fees and partner API costs) are required by the proposed stack but cannot be priced yet. They are not included in the totals above, so the monthly operating cost and the break-even figures are a lower bound until these are estimated.

Revenue. The revenue model is adaptive: the main source of income depends on the stage of the platform. Early on, income comes mainly from listing promotion and service commissions; subscriptions and advertising grow with the user base. Table 10.9 gives the unit values used in the break-even analysis. Advertising is excluded from the break-even calculation to keep it conservative.

Revenue stream	Description	Unit value
Listing and boosting	Providers pay to promote a listing	150 per listing per month
Pro subscription	Providers pay for advanced management tools	300 per provider per month
Provided services	Commission on each completed delivery or roadside request	25 per request
Advertisement	Sponsored placements by parts brands and service partners	Excluded from break-even

Table 10.9: Revenue assumptions (L.E)

Paying providers are assumed to be split equally between boosting and subscription, giving an average of L.E 225 per paying provider per month.

Break-even analysis. Monthly operating cost, including contingency, is L.E 17,236. Table 10.10 compares three scenarios against this cost, and Figure 1.2 shows the same comparison.

Measure	Conservative	Base	Optimistic
Paying providers	40	80	150
Completed requests	200	400	800
Revenue (L.E)	14,000	28,000	53,750
Operating cost (L.E)	17,236	17,236	17,236
Net result (L.E)	−3,236	+10,764	+36,514
Payback of one-off cost	Not reached	About 3.2 months	About 0.9 months

Table 10.10: Break-even scenarios (monthly, L.E)

In the base case, monthly revenue of L.E 28,000 exceeds operating cost by L.E 10,764, which recovers the one-off cost of L.E 34,430 in about 3.2 months once that level is reached. With 400 completed service requests per month, equal to about 2.8% of the 14,200 obtainable vehicles, the platform breaks even with 33 paying providers. Without any service revenue, it needs 77 paying providers. If 5% of registered providers pay, 77 paying providers require about 1,540 registered providers; at 10% they require about 770. These figures will be compared with the provider count estimated in Section 10.2.

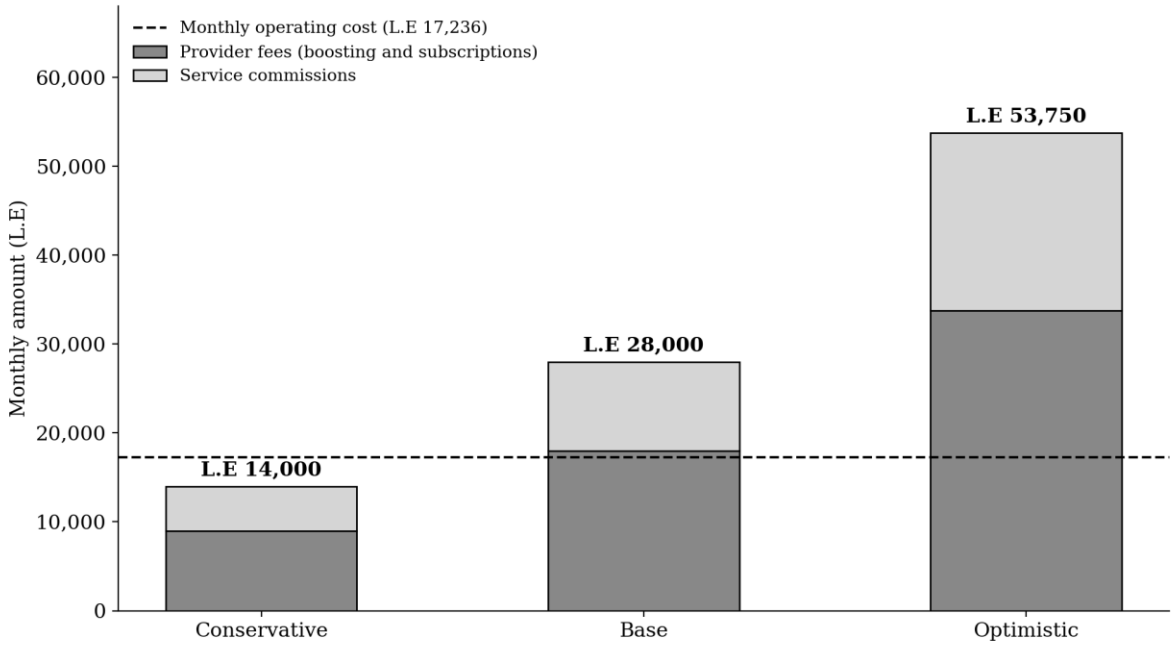


Figure 10.2: Monthly revenue by scenario against operating cost

Limitations and non-financial benefits. The model depends on assumed prices, conversion rates and request volumes, which the provider survey must test. Advertising income is excluded, and team labour is not costed. Beyond revenue, the platform offers drivers time savings and price transparency, and gives small providers a low-cost channel to reach customers.

Economic feasibility rating: feasible with conditions. Costs are low at prototype scale and break-even needs a modest number of paying providers. The rating depends on enough providers being willing to pay and on cloud, map and AI costs staying within budget.

10.6 Organizational Feasibility

Organizational feasibility asks whether the project team has the structure, commitment, authority and resources needed to deliver Tsl7tk and support its continuation. Operational feasibility addresses daily workflows; organizational feasibility assesses whether the team behind those workflows is ready to take responsibility for them.

Team structure and ownership. The proposed five-member team will assign ownership of mobile development, back end and data, the DSS knowledge base, interface design and testing, consistent with Section 10.3. A project coordinator should be designated from the team to track progress and dependencies. Named owners, available working hours and any skills gaps remain to be confirmed before implementation.

Decision-making and support. The team should agree how requirements, scope changes and releases are approved, and review progress regularly with the academic supervisor. Mechanic participation is needed to validate diagnostic rules, while provider participation is needed for pilot onboarding. These commitments are proposed requirements, not confirmed agreements.

Resources and continuity. Organizational readiness depends on access to development tools, test devices, shared documentation and an approved pilot budget. Training should address confirmed skills gaps. Before public launch, the team must identify who will own administration, maintenance and service accounts after the student project ends, and prepare a documented handover.

Area	Requirement	Planned action
Project alignment	A shared scope consistent with the project objectives	Agree the minimum viable product and confirm priorities with the academic supervisor
Team responsibilities	Clear ownership across the five-member team	Assign named module owners and a project coordinator; document responsibilities
Capacity and skills	Enough time and relevant skills to deliver the agreed scope	Confirm working hours and skills; arrange targeted training and reduce scope if needed
Decision-making	A clear process for approvals and scope changes	Agree who approves requirements, changes and releases; record decisions
Stakeholder support	Supervisor guidance and access to mechanics and providers	Confirm review arrangements, mechanic validation support and pilot provider participation
Resources and budget	Access to tools, test devices, accounts and pilot funding	Confirm resource availability, budget ownership and spending approval before commitments
Coordination and documentation	Shared information and regular progress reviews	Use the proposed GitHub and Notion tools; track tasks, dependencies and handover notes
Continuity and ownership	Responsibility for maintenance after the student project	Identify the future owner and support team; document account transfer and maintenance duties

Table 10.11: Organizational requirements and planned actions

Organizational feasibility rating: feasible with conditions. A five-member team offers a basis for organizing the prototype, but readiness is not yet demonstrated. The rating depends on confirming individual responsibilities, time availability, supervisor and external support, resource access and ownership after the student project.

10.7 Risk Assessment

Table 10.12 lists the main project risks. Likelihood and impact are scored Low (1), Medium (2) or High (3), and the score is their product. Scores of 6 or more are treated as high risks that need active management.

ID	Risk	Likelihood	Impact	Score	Mitigation
R1	Low provider adoption	High (3)	High (3)	9	Direct registration in Cairo, free basic listing, onboarding support, request leads
R2	Outdated listings, prices or stock	High (3)	Medium (2)	6	Update reminders, outdated-listing flag, administrator review
R3	Fake or abusive reviews	Medium (2)	High (3)	6	Reviews only for completed requests, abuse reporting, moderation
R4	Incorrect chatbot or DSS advice	Medium (2)	High (3)	6	Mechanic-validated rules, limited fault categories, advisory wording
R5	Data-protection non-compliance	Medium (2)	High (3)	6	Consent records, minimal data, data-transfer assessment (including hosting region), legal review before launch

ID	Risk	Likelihood	Impact	Score	Mitigation
R6	Scope too large for a team of five	High (3)	High (3)	9	Build in tiers: MVP first (accounts, garage, search, requests, reviews, admin), then chatbot/DSS, then roadside and delivery as integrations; assign module owners, review scope at milestones
R7	Cloud, map or AI costs exceed budget	Medium (2)	Medium (2)	4	Usage budgets and alerts, caching, cost review before adopting paid services
R8	Providers bypass the platform to avoid fees	Medium (2)	Medium (2)	4	Charge for visible value (leads, verification, promotion) rather than blocking contact
R9	Incomplete or outdated provider data	Medium (2)	Medium (2)	4	Supplement Google Maps data with direct registration, bulk CSV / Excel upload and provider self-updates; label demonstration records
R10	Delivery or roadside partners unreliable	Medium (2)	Medium (2)	4	Partner terms, status tracking, fallback to listed providers
R11	Exchange rate rise increases foreign-priced costs	Medium (2)	Low (1)	2	Contingency allowance, regular cost review

Table 10.12: Risk register

The two highest risks are low provider adoption (R1) and a scope that is large for a team of five (R6). Both are addressed by building the core modules first and by securing an early provider base through direct registration.

10.8 Feasibility Conclusion

Dimension	Rating	Key condition
Market	Feasible with conditions	The survey confirms demand and the gaps in Related Work are real
Technical	Feasible with conditions	Map, SMS and LLM costs kept within budget; DSS limited to validated fault categories; provider data quality secured
Operational	Feasible with conditions	Enough providers register and keep listings current; moderation capacity grows with usage
Economic	Feasible with conditions	At least 33 to 77 paying providers, depending on service revenue; costs kept within budget
Organizational	Feasible with conditions	Confirm roles, time availability, stakeholder support, resources and ownership after the student project
Overall	Feasible with conditions	All the above

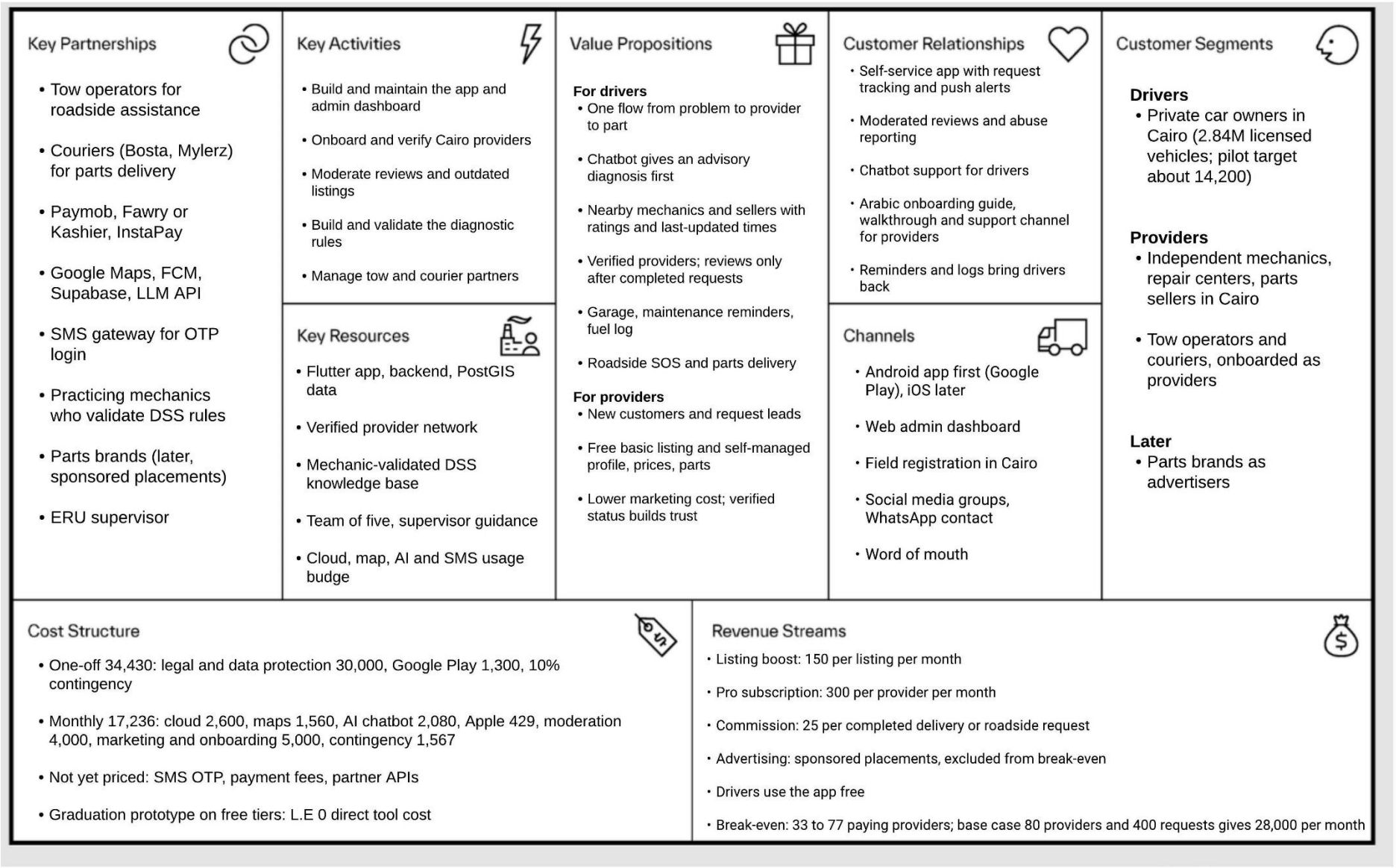
Table 10.13: Summary of feasibility ratings

Overall conclusion. Tsl7tk is feasible with conditions. The vehicle market is large and growing, the technology is mature and affordable at prototype scale, the operating processes are simple, and break-even needs a modest number of paying providers. The main risks are provider adoption, the size of the scope and the accuracy of the diagnostic chatbot. Organizational readiness also depends on confirmed responsibilities, resources and continuity arrangements.

Recommendations. The project should proceed, and should be subject to the following actions:

- Run the driver and provider survey to confirm demand and willingness to pay.
- Register an initial base of Cairo providers directly before public launch.
- Limit the diagnostic chatbot to fault categories validated by practicing mechanics.
- Complete a legal review and data-protection set-up before collecting real user data.
- Confirm team responsibilities, available time, stakeholder support and ownership after the student project.
- Set usage budgets for cloud, map, SMS and AI services and review costs regularly.

10. Business Model



11. Go-To-Market Strategy

11.1 GTM Summary

TSL7TK is a two-sided platform: **drivers** need a reliable way to go from a vehicle problem to a provider and a part, and **providers** (mechanics, repair centers, parts sellers) need customers at low cost. A marketplace is worthless on day one if either side is empty, so the whole strategy is built around one rule: **win supply first, in dense pockets of Cairo, then pull in demand.**

The plan in one paragraph: validate demand with a survey and count Cairo providers; seed a verified base of providers through direct field registration and assisted onboarding; launch to drivers on Android in a few pilot zones using social groups, WhatsApp sharing and partner workshops; keep everything free until there is real density and completed requests; then introduce boosts, Pro subscriptions and service commissions; finally add the chatbot/DSS, roadside and delivery as partner integrations and expand beyond Cairo.

North-star metric: completed requests per month. The base-case break-even scenario in the feasibility study needs about 400 per month with 80 paying providers.

11.2 Two Tracks: Graduation Prototype and Public Pilot

The proposal separates a free, emulator-based graduation prototype (L.E 0 direct tool cost) from a later paid public pilot (one-off L.E 34,430 and about L.E 17,236 per month, excluding SMS, payment fees and partner API costs). The GTM follows the same split.

- **Track A: Validation GTM (graduation prototype, zero budget).** Goal: prove demand and provider willingness, not revenue. Activities: driver and provider survey, mechanic interviews, provider counting, clickable Figma prototype tests, demo records clearly labelled as demo.

- **Track B: Public pilot GTM (after graduation, funded).** Goal: reach the break-even range of 33 to 77 paying providers. Activities: everything in Sections 6 to 10.

Track B should start only if Track A meets its gates (Section 11).

11.3 Target Market and Segmentation

Market size (from the feasibility study): 11.05 million licensed vehicles in Egypt, 2.84 million in Cairo (25.7%), and an obtainable pilot market of about 14,200 vehicles *(assumption: 0.5% of Cairo vehicles reached)*.

Drivers (demand side)

- **Core segment:** private car owners in Cairo who need maintenance, repair or parts.
- **Beachhead hypothesis to test in the survey:** owners of older or high-mileage cars (including people who drive for work), because they visit mechanics and parts shops most often and feel fuel and repair costs most. This is a hypothesis, not a finding.
- **Needs we serve** finding a trusted mechanic nearby, comparing parts availability and price, understanding a fault before visiting a shop, getting help in a breakdown, and keeping maintenance and fuel records.

Providers (supply side)

- **Parts sellers and independent mechanics/repair centers** operating in Cairo. They are the priority because drivers cannot get value without them.
- **Tow operators and couriers**, onboard as partners/providers (not employees), for roadside assistance and parts delivery.
- **Later:** parts brands and service partners as sponsored advertisers.

Where to start geographically

Do not launch "Cairo" evenly. Using Google Maps place counts already planned in the feasibility study to find clusters with many workshops and parts shops close together and launch in 3 to 5 pilot zones. A driver who searches in a zone with ten verified providers will trust the app; one who sees two will not return.

11.4 Value Proposition and Positioning

Positioning statement: For car owners in Cairo who lose time and money searching for mechanics and parts, TSL7TK is the one app that takes you from the problem to a verified provider and the right part, with request tracking, moderated reviews and a maintenance and fuel log. Unlike parts-only or discovery-only apps, it covers the whole repair journey.

Messaging pillars

For drivers

- **One flows from problem to part.** Describe the problem, see nearby verified providers, request the part or service.
- **Know before you go.** An advisory diagnosis from the chatbot, always confirmed by a provider (never presented as a final diagnosis).
- **Trust you can check.** Verified providers, reviews only after completed requests, and visible last-updated times on prices and stock.
- **Help when it breaks down.** Roadside SOS and parts delivery in the same app.
- **Save money and time.** The proposal estimates about EGP 6,300 per driver per year plus around 20 hours saved. Present this as an estimate until the survey refines it.

For providers

- **New customers and requests leads at low cost.** Free basic listing, direct requests.
- **A verified badge** that builds trust without paying for advertising.
- **Simple tools.** Manage profile, prices and parts from one app, with an Arabic onboarding guide.
- **Estimated value:** about EGP 42,000 per provider per year (EGP 36,000 extra revenue plus EGP 6,000 marketing savings). Treat as an estimate to be validated, and do not promise it in marketing.

Competitive framing

The related-work analysis groups competitors into parts marketplaces (Ringaat, Dynamu, EgyCarParts, N Auto Express), service and discovery apps (Sayyn, Dawar, Engazny) and general marketplaces (3arabawy). The chapter itself notes that missing features are "not documented", not proven absent, so the GTM should **never claim competitors lack a feature** until the apps have been tested directly.

Competitor type	Their strength	Our message
Parts marketplaces (Ringaat, Dynamu)	Large catalogues, quotations, delivery	You do not need to know the part. Start from the problem.
Service apps (Sayyn)	Booking, towing, maintenance records	Parts, providers and diagnosis in one place.
Discovery apps (Dawar, Engazny)	Nearby shops on a map	Request tracking and moderated, request-linked reviews.
Free alternatives (Facebook groups, maps, word of mouth)	Free and familiar	Same convenience, with verification and accountability.

Stance: TSL7TK does not try to beat anyone at catalogue depth or tow-network size. It wins on integration and trust.

11.5 Business Model and Pricing Strategy

From the feasibility study:

- **Drivers:** free to use.
- **Listing boost:** EGP 150 per listing per month.
- **Pro subscription:** EGP 300 per provider per month.
- **Commission:** EGP 25 per completed delivery or roadside request.
- **Advertising:** sponsored placements by parts brands, excluded from break-even.
- **Average revenue per paying provider:** EGP 225 per month (50/50 split of boost and subscription).

Pricing sequence

1. **Free phase (pilot launch until density targets are met):** free basic listing, free request leads, free verification. Purpose: remove every barrier to registering.
2. **Soft monetization:** once a zone shows repeat requests, offer paid boosts first. They are optional and visibly tied to results (more views, more requests).
3. **Subscription:** introduce Pro tools for providers who already receive requests through the app.
4. **Commissions:** start with delivery and roadside requests, where the platform clearly adds value through partners.
5. **Advertising:** only after a meaningful driver base exists.

Guardrails

- **Charge for visible value** (leads, verification, promotion) rather than blocking contact. This answers risk R8 (providers bypassing the platform to avoid fees).
- **Acquisition cost guideline:** keep the cost to acquire one paying provider below roughly two months of its revenue (about EGP 450 at EGP 225 per month). This is a rule of thumb for the team, not a figure from the proposal.

11.6 Phase 1: Supply Seeding (Solving the Cold Start)

The feasibility study already names the cold start (empty marketplace on day one) as a technical constraint. This is how GTM addresses it.

Targets

Break-even needs 33 to 77 paying providers. The study notes that at 5% paid conversion, 77 paying providers means about 1,540 registered providers, and 10% about 770. Those are large numbers for a student team. Practical approach:

- Set the pilot goal at **roughly 800 registered providers across the pilot zones *(assumption matches 10% conversion to 80 paying providers)***.
- Compare it with the real Cairo provider, once the Google Maps count is done and **scale the target down per zone if the market is smaller than expected**, rather than spreading thin.
- Quality beats count: 150 verified, responsive providers in the right zones are worth more than 800 empty profiles.

Tactics

- **Direct field registration.** The team visits workshops and parts shops in the pilot zones. Registration on the spot, using phone number OTP, takes minutes.
- **Assisted onboarding.** Many mechanics have limited familiarity with technology, so the team enters their first profile, services and first 10 to 20 parts together with them.
- **Bulk upload.** Parts sellers who keep stock in spreadsheets send the file and the team imports it (CSV/Excel upload is already in the plan).
- **Verification as an incentive.** Administrators check business details and mark the listing verified. Position this as free credibility, not a chore.
- **Arabic onboarding guide and short in-app walkthrough** (already in the operational plan), plus a WhatsApp support line for questions.
- **Mechanic champions.** Recruit a small group of practicing mechanics who also validate the DSS rules. They become early adopters, reference customers and word-of-mouth sources within their neighborhoods.
- **Roadside and delivery partners.** Sign tow operators and check courier API fit (Bosta, Mylerz) before promising SOS or delivery to drivers.

3 What not to do

- Do not scrape competitor listings (breaches terms, goes stale, and conflicts with the proposal's own data policy).
- Do not present demo records as real. They must be clearly labelled.
- Do not promise livestock levels. The platform tracks accept/decline responses, not inventory.

11.7 Phase 2: Driver Launch

Launch scope

- **Android first** (iOS later, as in the technology plan; the Apple developer fee of about L.E 5,148 per year can wait).
- **MVP features only:** accounts, garage, search, requests, reviews and administration. The chatbot/DSS and roadside/delivery follow in later tiers, matching the risk mitigation for R6 (scope too large).
- **Pilot zones only.** Marketing is targeted at the neighborhoods where supply is dense.

Channels

Following the channels in the business model canvas (Android app, field registration, social media groups, WhatsApp contact, word of mouth):

- **Closed beta with ERU community.** Students, staff and families with cars give the first real usage data and early reviews in a friendly setting.
- **Social media groups.** Car-owners and neighborhood groups are both a competitor and the best channel: they are where drivers already ask, "who fixes this?". Participate with helpful content and let the app answer, rather than spamming.
- **WhatsApp sharing.** Because sellers and mechanics already communicate through WhatsApp, make it easy to share a provider profile or request status as a link.
- **QR codes at partner workshops and shops.** Verified providers display a sticker and invite their own customers to install the app, leave a review after a completed request and log maintenance.
- **Content marketing.** Short Arabic posts and videos: what a noise or warning light may mean (from the validated fault categories), maintenance schedules and fuel-saving tips. This also showcases the chatbot before it launches.
- **Referring to incentives.** Small, low-cost rewards for referrals, funded from the marketing budget.
- **Word of mouth.** The most trusted channel in this market and the one the app is trying to formalize. Reviews linked to completed requests turn it into something verifiable.

Retention hooks

The garage, maintenance reminders and fuel log are the reasons for a driver coming back when nothing is broken. Make setting up the first vehicle part of onboarding and send reminders through push notifications (Firebase Cloud Messaging, no cost).

11.8 Phase 3: Monetization and Optimization

- Trigger for introducing paid features: a zone has repeat requests, an acceptable provider response rate and early reviews (see gates in Section 11).
- Offer boosts to providers already receiving leads. Show them their own numbers (views, requests, completed jobs).
- Test pricing with a handful of providers before rolling out. Keep the proposal's unit values (EGP 150, 300, 25) as the starting point and let the survey and early conversion decide.
- Run the break-even scenarios as a live dashboard: paying providers, completed requests, revenue against the L.E 17,236 monthly operating cost (lower bound until SMS, payment and partner API costs are priced).

11.9 Phase 4: Expansion

1. **Chatbot and DSS launch** with only the mechanic-validated fault categories, and always advisory.
2. **Roadside and parts delivery** through partner integrations, including commission revenue.
3. **iOS release.**
4. **Zone expansion within Cairo**, then other governorates only if the pilot proves successful (as stated in the constraints).
5. **Sponsored placements** by parts brands.

11.10 Marketing Budget

The cost structure allocates L.E 5,000 per month to marketing and provider onboarding. A suggested split *(assumption, adjust after the first month)*:

Item	Monthly (L.E)
Field onboarding costs (transport, stipends, printed guides)	2,000
Community content and social posts	1,200
Driver referral rewards	1,000
QR stickers and printed materials for partner shops	500
Contingency	300
Total	**5,000**

Track A (the graduation prototype) assumes no paid marketing. Surveys, interviews and

prototype tests use free tools only.

10.11 KPIs and Stage Gates

Stage	Key metrics	Gate to move on *(proposed)*
Validation	Survey responses; share of drivers who would use the app; share of providers willing to join or pay; Cairo provider count	Survey confirms demand and the gaps in related work are real (the study's own condition)
Supply seeding	Registered, verified and active providers per zone; parts listed per provider; time to complete onboarding	Enough verified providers in each pilot zone for a driver search to return useful results
Driver launch	Installs, vehicles added to garage, weekly active users, requests per week, request acceptance rate, completion rate, reviews per completed request	Repeat requests and healthy acceptance rate in at least some zones
Monetization	Paying providers, revenue per paying provider (target EGP 225), conversion from free to paid, commissions	Moving toward 33 to 77 paying providers
Expansion	Completed requests per month (base case 400), chatbot usage, SOS and delivery requests, DSS feedback	Revenue above the monthly operating cost

Set numeric thresholds for the gates once the survey and the provider count are available.

Avoid inventing them before there is data.

10.12 Trust, Quality and Compliance

Trust is the product, so it is part of the GTM and not only a technical feature.

- **Verification** by administrators before a listing is marked verified.
- **Reviews only for completed requests**, with abuse reporting and moderation, to resist fake reviews (R3).
- **Update timestamps** on prices and stock; flag outdated listings (R2).
- **Advisory wording** on all chatbot and DSS output, limited to validated fault categories (R4).
- **Data protection:** complete the legal review and data-protection set-up before collecting real user data, in line with Egypt's Personal Data Protection Law No. 151 of 2020 and its executive regulations cited in the references (R5).
- **Honest marketing:** use estimates as estimates, label demo data, and make no claims about competitors that have not been tested.

10.13 Risks Specific to Go-to-Market

Risk	Link to proposal	Response
Providers do not adopt or keep listings current	R1, R2	Free basic listing, assisted onboarding, update reminders, outdated-listing flag, early request leads
Scope too large for five students	R6	Launch MVP only; chatbot/DSS and roadside/delivery as later tiers; review scope at milestones
Providers bypass the platform to avoid fees	R8	Charge for leads, verification and promotion, not for contact
Free alternatives (social groups, maps) are good enough	Threats in SWOT	Competing on reliability: verification, tracking, request-linked reviews
Roadside or delivery partners underperform	R10	Partner terms, status tracking, fallback to listed providers; launch these services only after partners are confirmed
Costs exceed budget	R7	Usage budgets and alerts, deep links instead of the Directions API, cost review before paid services
Weak or fake reviews hurt trust	R3	Reviews linked to completed requests, moderation queue

10.14 Open Items to Confirm

These gaps in the proposal affect the GTM and should be filled before Track B:

1. **Number of providers in Cairo** (the placeholder in Section 10.2), which sets the realistic supply target.
2. **Survey results** on driver demand, provider willingness to join and willingness to pay.
3. **Prices of SMS OTP, payment gateway fees and courier/tow partner terms**, which are not yet in the cost totals.
4. **List of validated fault categories** for the first DSS release (placeholder in Section 10.3).
5. **Team skills and roles** (placeholder in Section 10.3), including who owns field onboarding, content and partnerships.
6. **Sponsor and long-term owner** after the student project, which determines whether Track B can happen at all.

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